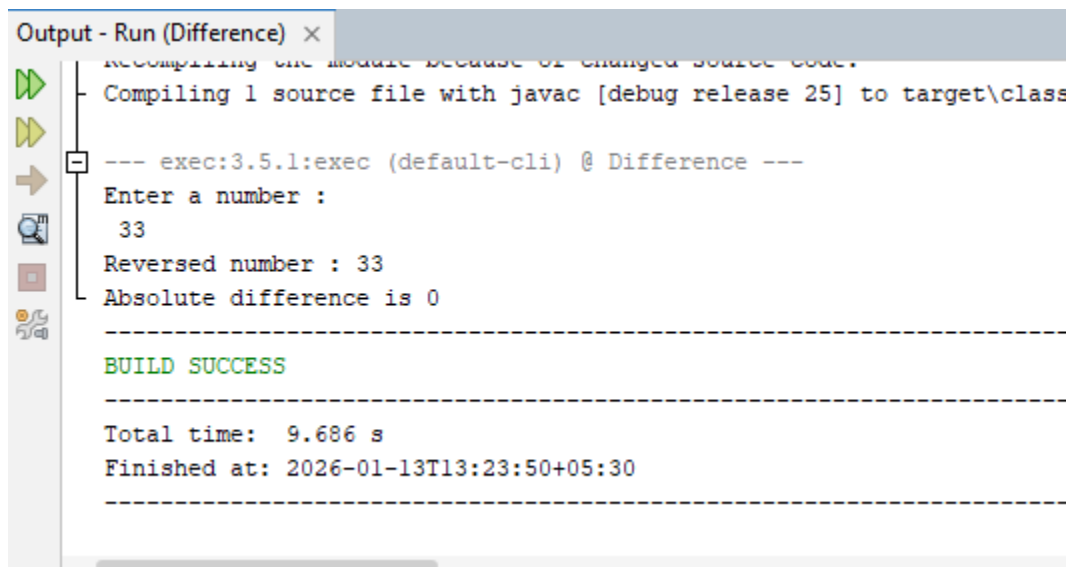


PROGRAM 1

```
import java.util.*;
class Difference{
public static void main(String args[]){
Scanner sc = new Scanner(System.in);
System.out.println("Enter a number :"); int
num = sc.nextInt();
int rev = 0,dup = num;
while(dup > 0){
int d = dup % 10;
rev = rev*10 + d;
dup = dup/10; }
int dif = Math.abs(num - rev);
System.out.println("Reversed number : "+rev);
System.out.println("Absolute difference is " +dif);
}
}
```

OUTPUT



The screenshot shows an IDE's output window titled "Output - Run (Difference) x". The output text is as follows:

```
Recompiling the module because of changed source code.
Compiling 1 source file with javac [debug release 25] to target\class
--- exec:3.5.1:exec (default-cli) @ Difference ---
Enter a number :
33
Reversed number : 33
Absolute difference is 0

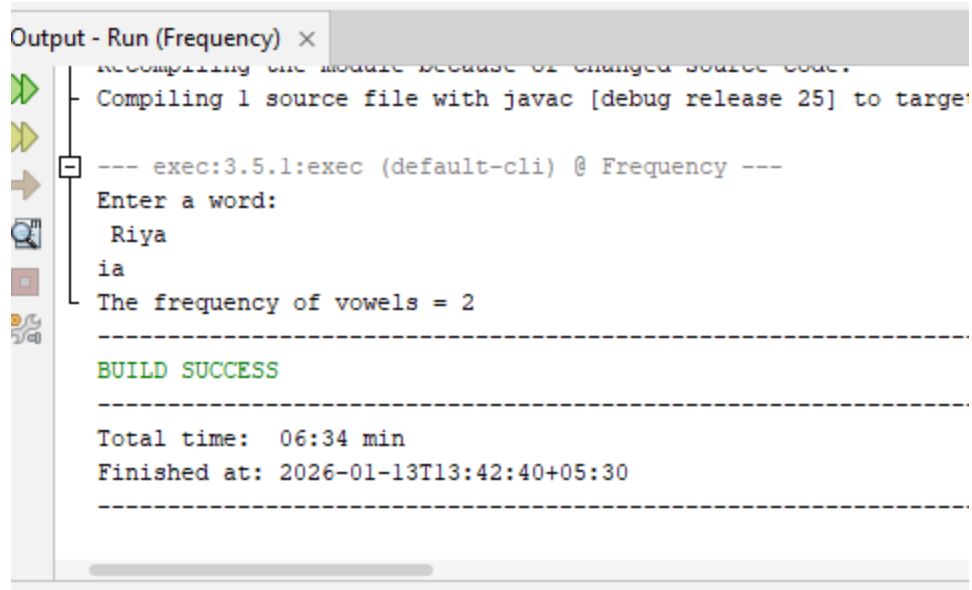
-----
BUILD SUCCESS
-----

Total time: 9.686 s
Finished at: 2026-01-13T13:23:50+05:30
-----
```

PROGRAM 2

```
import java.util.*;
class Frequency{
public static void main(String args[]){
Scanner sc = new Scanner(System.in);
System.out.println("Enter a word:");
String s = sc.next();
int l = s.length();
int c = 0;
for(int i = 0; i < l; i++){
char ch = s.charAt(i);
if("AEIOUaeiou".indexOf(ch) != -1){
System.out.print(ch);
c = c+1;
}
}
System.out.println();
System.out.println("The frequency of vowels = "+ c);
}
}
```

OUTPUT



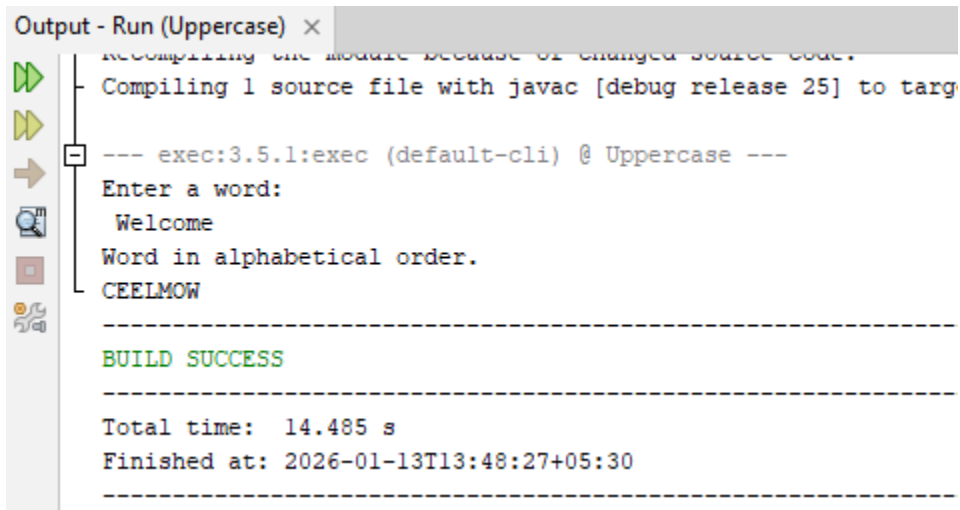
```
Output - Run (Frequency) ×
Recompiling the module because of changed source code.
Compiling 1 source file with javac [debug release 25] to target
--- exec:3.5.1:exec (default-cli) @ Frequency ---
Enter a word:
Riya
ia
The frequency of vowels = 2
BUILD SUCCESS
Total time: 06:34 min
Finished at: 2026-01-13T13:42:40+05:30
```

PROGRAM 3

```
import java.util.*;
class Uppercase {
public static void main(String args[]){
Scanner sc = new Scanner(System.in);
System.out.println("Enter a word:");
String s = sc.next();
s = s.toUpperCase();
int l = s.length();
char ch;
System.out.println("Word in alphabetical order.");
for(int i = 65; i <= 90; i++){
for(int j = 0; j < l; j++){
ch = s.charAt(j);
if(ch == (char)i){
System.out.print(ch);
}
}
}
}
```

```
}
```

OUTPUT



```
Output - Run (Uppercase) x
Recompiling the module because of changed source code.
Compiling 1 source file with javac [debug release 25] to targ
--- exec:3.5.1:exec (default-cli) @ Uppercase ---
Enter a word:
Welcome
Word in alphabetical order.
CEELMOW
-----
BUILD SUCCESS
-----
Total time: 14.485 s
Finished at: 2026-01-13T13:48:27+05:30
-----
```

PROGRAM 4

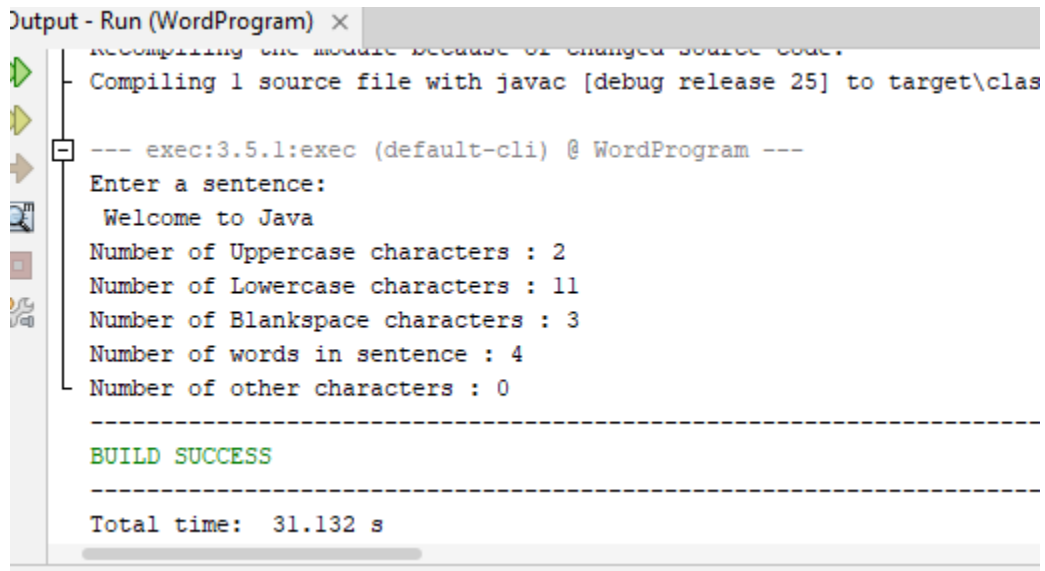
```
import java.util.*;
class WordProgram {
public static void main(String args[]){
Scanner sc = new Scanner(System.in);
System.out.println("Enter a sentence:");
String s = sc.nextLine();
int u = 0,l = 0,b = 0,w = 1,o = 0; int
len = s.length();
for(int i = 0; i < len; i++){
char ch = s.charAt(i);
if(Character.isUpperCase(ch)){
u = u+1;
}else if(Character.isLowerCase(ch)){
l = l+1;
}else if(Character.isWhitespace(ch)){
b = b+1;
w = w+1;
```

```

}else{
o = o+1;
}
}
System.out.println("Number of Uppercase characters : "+u);
System.out.println("Number of Lowercase characters : "+l);
System.out.println("Number of Blankspace characters : "+b);
System.out.println("Number of words in sentence : "+w);
System.out.println("Number of other characters : "+o);
}
}

```

OUTPUT



```

Output - Run (WordProgram) x
Recompiling the module because of changed source code.
Compiling 1 source file with javac [debug release 25] to target\clas
--- exec:3.5.1:exec (default-cli) @ WordProgram ---
Enter a sentence:
Welcome to Java
Number of Uppercase characters : 2
Number of Lowercase characters : 11
Number of Blankspace characters : 3
Number of words in sentence : 4
Number of other characters : 0
-----
BUILD SUCCESS
-----
Total time: 31.132 s

```

PROGRAM 5

```

import java.util.*;
class Palindrome {
public static void main(String args[]){
Scanner sc = new Scanner(System.in);
System.out.println("Enter a word :");

```

```

String s = sc.next();
char ch ;
String s1 = "";
int l =s.length();
for(int i =0; i < l; i++){
    ch = s.charAt(i);
    s1 = ch + s1;
}
if(s.equals(s1)){
    System.out.println("The word is a palindrome.");
}else{
    System.out.println("The word is not a palindrome.");
    String s2 = s + s1;
    System.out.println("Word changed to palindrome is : "+ s2);
}
}
}
}

```

OUTPUT

```

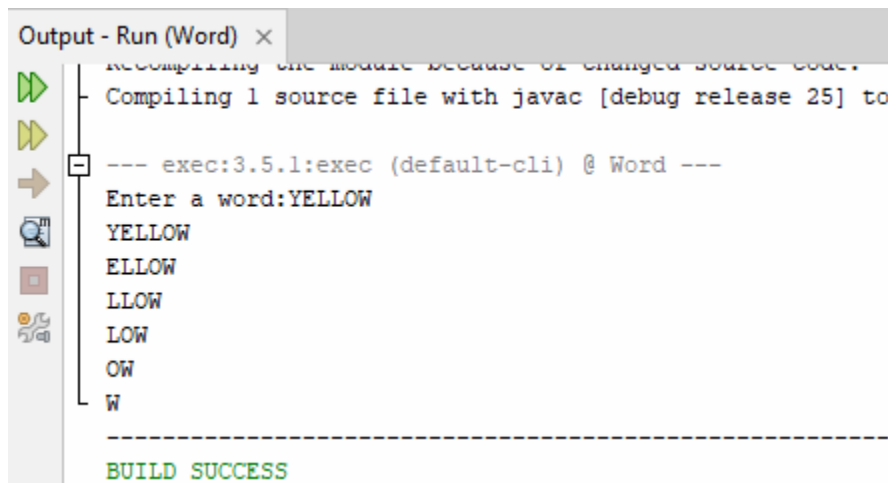
Output - Run (Palindrome) x
--- exec:3.5.1:exec (default-cli) @ Palindrome ---
Enter a word :
    Madam
The word is not a palindrome.
Word changed to palindrome is : MadammadaM
-----
BUILD SUCCESS
-----
Total time:  14.312 s
Finished at: 2026-01-13T14:11:26+05:30
-----

```

PROGRAM 6 :

```
import java.util.Scanner;
public class Word
{
    public static void main(String args[]) {
        Scanner in = new Scanner(System.in);
        System.out.print("Enter a word:");
        String word = in.nextLine();
        int len = word.length();
        for (int i = 0; i<len; i++) {
            for (int j = i; j<len; j++) {
                char ch = word.charAt(j);
                System.out.print(ch); }
            System.out.println();
        }
    }
}
```

OUTPUT:



```
Output - Run (Word) x
Recompiling the module because of changed source code.
Compiling 1 source file with javac [debug release 25] to
--- exec:3.5.1:exec (default-cli) @ Word ---
Enter a word:YELLOW
YELLOW
ELLOW
LLOW
LOW
OW
W
-----
BUILD SUCCESS
```

PROGRAM 7:

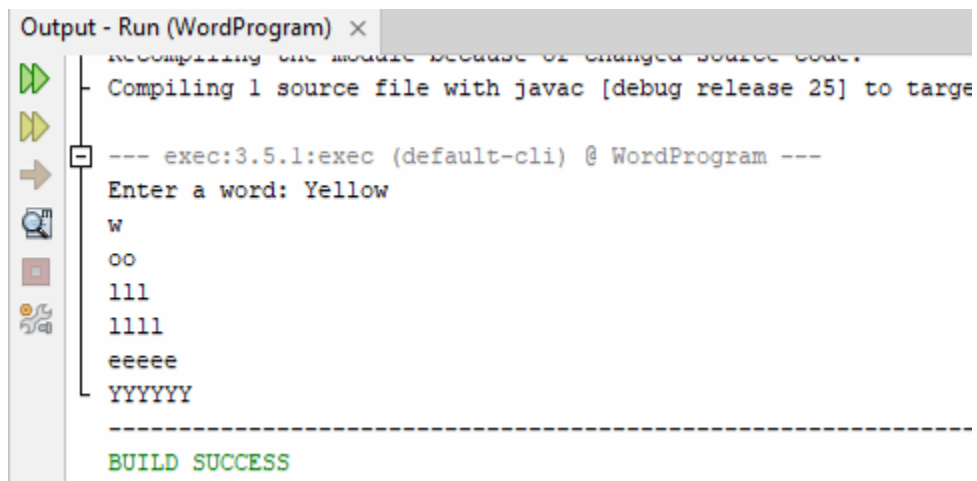
```
import java.util.Scanner;
```

```

public class WordProgram
{
    public static void main(String args[])
    {
        Scanner in = new Scanner(System.in);
        System.out.print("Enter a word: ");
        String word = in.nextLine();
        int len = word.length();
        for (int i = len - 1; i >= 0; i--)
        {
            for (int j = len - 1; j >= i; j--)
            {
                char ch = word.charAt(i);
                System.out.print(ch);
            }
            System.out.println();
        }
    }
}

```

OUTPUT:



```

Output - Run (WordProgram) x
--- exec:3.5.1:exec (default-cli) @ WordProgram ---
Enter a word: Yellow
w
oo
lll
llll
eeee
YYYYYY
-----
BUILD SUCCESS

```

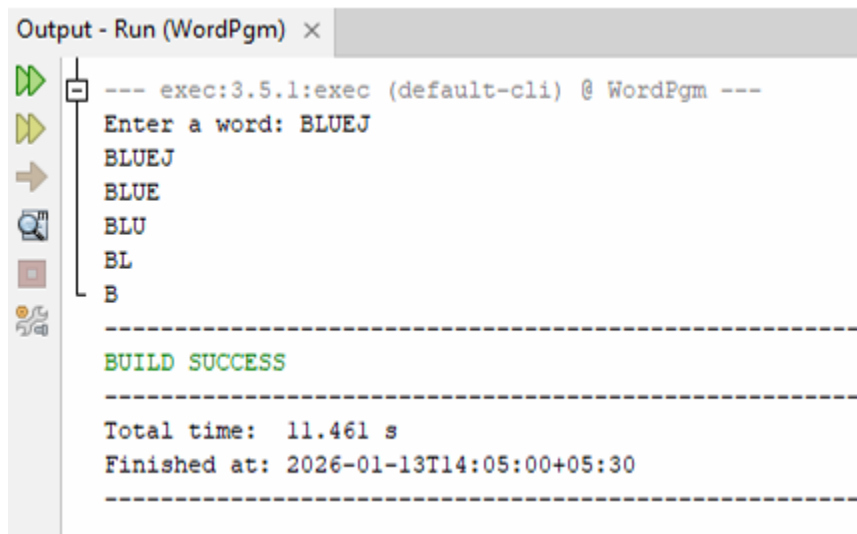
PROGRAM 8:


```

import java.util.Scanner;
public class WordPgm {
    public static void main(String args[]) {
        Scanner in = new Scanner(System.in);
        System.out.print("Enter a word: ");
        String word = in.nextLine();
        int len = word.length();
        for (int i = len; i > 0; i--) {
            for (int j = 0; j < i; j++) {
                System.out.print(word.charAt(j));
            }
            System.out.println();
        }
    }
}

```

OUTPUT:



```

Output - Run (WordPgm) x
--- exec:3.5.1:exec (default-cli) @ WordPgm ---
Enter a word: BLUEJ
BLUEJ
BLUE
BLU
BL
B
-----
BUILD SUCCESS
-----
Total time: 11.461 s
Finished at: 2026-01-13T14:05:00+05:30
-----

```

PROGRAM 9:

```

import java.util.Scanner;
public class KboatSDANames
{

```

```

public static void main(String args[]) {
    Scanner in = new Scanner(System.in);
    String names[] = new String[10];
    System.out.println("Enter 10 names");
    for (int i = 0; i<names.length; i++) {
        names[i] = in.nextLine();
    }
    System.out.print("Enter a letter:");
    char ch = in.next().charAt(0);
    ch = Character.toUpperCase(ch);
    for (int i = 0; i<names.length; i++) {
        if (Character.toUpperCase(names[i].charAt(0)) == ch) {
            System.out.println(names[i]);
        }
    }
}

```

OUTPUT:



```

Output - Run (KboatSDANames) x
--- exec:3.5.1:exec (default-cil) @ KboatSDANames ---
Enter 10 names
Hema
Dharshana
Boomika
Kanishga
Harini
Idhayahasini
Kaavika
Dharuni Sri
Gayathri
Varshika
Enter a letter:H
Hema
Harini

```

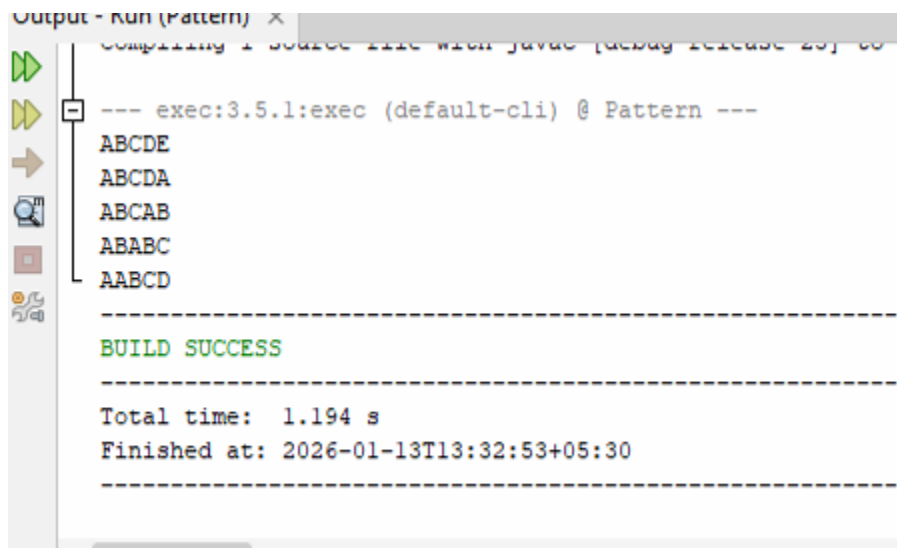
PROGRAM 10:

```

public class Pattern
{
    public static void main(String args[]) {
        String word = "ABCDE";
        int len = word.length();
        for (int i = 0; i<len; i++) {
            for (int j = 0; j<len - i; j++) {
                System.out.print(word.charAt(j));
            }
            for (int k = 0; k<i; k++) {
                System.out.print(word.charAt(k));
            }
            System.out.println();
        }
    }
}

```

OUTPUT :



```

Output - Run (Pattern) x
Compiling 1 source file with javac [debug release 20] ...
--- exec:3.5.1:exec (default-cli) @ Pattern ---
ABCDE
ABCD A
ABCAB
ABA BC
AABCD
-----
BUILD SUCCESS
-----
Total time: 1.194 s
Finished at: 2026-01-13T13:32:53+05:30
-----

```

PROGRAM 11:

```
import java.util.Scanner;

public class KboatSDANames
{
    public static void main(String args[]) {
        Scanner in = new Scanner(System.in);
        String names[] = new String[10];
        System.out.println("Enter 10 names");
        for (int i = 0; i<names.length; i++) {
            names[i] = in.nextLine();
        }
        System.out.print("Enter a letter:");
        char ch = in.next().charAt(0);
        ch = Character.toUpperCase(ch);
        for (int i = 0; i<names.length; i++) {
            if (Character.toUpperCase(names[i].charAt(0)) == ch) {
                System.out.println(names[i]);
            }
        }
    }
}
```

OUTPUT:

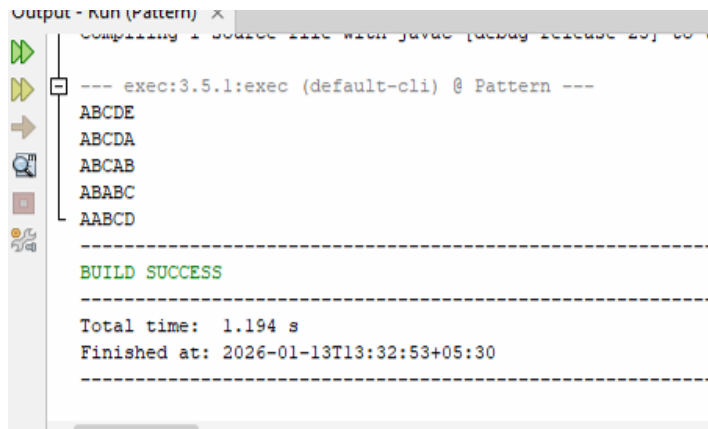
```
Output - Run (KboatSDANames) x
--- exec13.5.1:exec (default-cil) @ KboatSDANames ---
Enter 10 names
Hema
Dharshana
Boomika
Kanishga
Harini
Idhayahasini
Kaavika
Dharuni Sri
Gayathri
Varshika
Enter a letter:H
Hema
Harini
```

PROGRAM 12:

```
public class Pattern
public class Pattern{
public static void main(String args[]) {
String word = "ABCDE";
int len = word.length();
for (int i = 0; i<len; i++) {
for (int j = 0; j<len - i; j++) {
System.out.print(word.charAt(j));
}
for (int k = 0; k<i; k++) {
System.out.print(word.charAt(k));
}
System.out.println();
}
}
```

}

OUTPUT :



```
Output - Run (Pattern) x
Compiling 1 source file with javac [debug release 20] ...
--- exec:3.5.1:exec (default-cli) @ Pattern ---
ABCDE
ABCDA
ABCAB
ABABC
AABCD
-----
BUILD SUCCESS
-----
Total time: 1.194 s
Finished at: 2026-01-13T13:32:53+05:30
-----
```

PROGRAM 13:

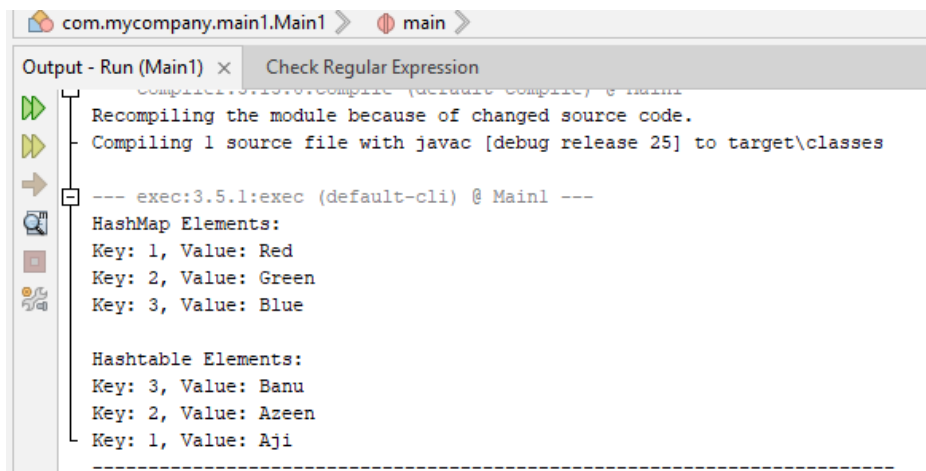
```
import java.util.*;
public class Main1 {
    public static void main(String[] args) {
        HashMap<Integer, String> hashMap = new HashMap<>();
        hashMap.put(1, "Red");
        hashMap.put(2, "Green");
        hashMap.put(3, "Blue");
        System.out.println("HashMap Elements: ");
        for (Integer key : hashMap.keySet()) {
            System.out.println("Key: " + key + ", Value: " + hashMap.get(key));
        }
        Hashtable<Integer, String> hashtable = new Hashtable<>();
        hashtable.put(1, "Aji");
        hashtable.put(2, "Azeen");
        hashtable.put(3, "Banu");
        System.out.println("\nHashtable Elements:");
    }
}
```

```

for (Integer key : hashtable.keySet()) {
    System.out.println("Key: " + key + ", Value: " + hashtable.get(key));
}
}
}

```

OUTPUT



```

com.mycompany.main1.Main1 > main >
Output - Run (Main1) x Check Regular Expression
--- exec:3.5.1:exec (default-cli) @ Main1 ---
Recompiling the module because of changed source code.
Compiling 1 source file with javac [debug release 25] to target\classes
HashMap Elements:
Key: 1, Value: Red
Key: 2, Value: Green
Key: 3, Value: Blue
Hashtable Elements:
Key: 3, Value: Banu
Key: 2, Value: Azeen
Key: 1, Value: Aji
-----

```

PROGRAM 14:

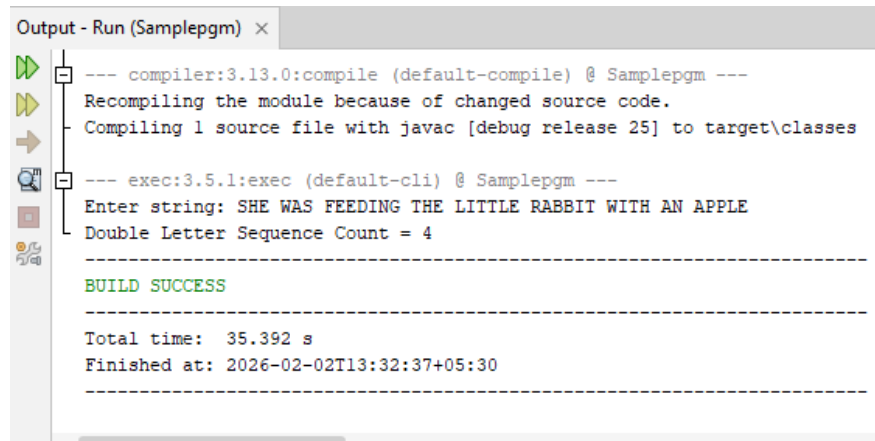
```

import java.util.Scanner;
public class Samplepgm {
    public static void main(String args[]) {
        Scanner in = new Scanner(System.in);
        System.out.print("Enter string: ");
        String s = in.nextLine();
        String str = s.toUpperCase();
        int count = 0;
        int len = str.length();
        for (int i = 0; i < len - 1; i++) {
            if (str.charAt(i) == str.charAt(i + 1))
                count++;
        }
    }
}

```

```
System.out.println("Double Letter Sequence Count = " + count);  
}  
}
```

OUTPUT:



The screenshot shows an IDE output window titled "Output - Run (Samplepgm) x". It contains the following text:

```
--- compiler:3.13.0:compile (default-compile) @ Samplepgm ---  
Recompiling the module because of changed source code.  
Compiling 1 source file with javac [debug release 25] to target\classes  
  
--- exec:3.5.1:exec (default-cli) @ Samplepgm ---  
Enter string: SHE WAS FEEDING THE LITTLE RABBIT WITH AN APPLE  
Double Letter Sequence Count = 4  
  
-----  
BUILD SUCCESS  
-----  
  
Total time: 35.392 s  
Finished at: 2026-02-02T13:32:37+05:30  
-----
```